

Exploring the Integration of Spatial Information into a Spectral Clustering Approach to Unsupervised Band Selection

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Abstract—Band selection methods are crucial for mitigating the challenge of high dimensionality in hyperspectral image data. Several unsupervised band selection techniques have already shown success, including the Particle Swarm Optimisation with Fuzzy Clustering Means (PSO-FCM). However, such an approach does not exploit the available spatial information. The work herein thus proposes Superpixel-enhanced PSO-FCM (SPPF), that exploits both the spectral and spatial redundancy in hyperspectral image data. To ensure more optimal exploitation of the local spatial context of the Superpixel map, various distance measures, together with various cluster-representative selection techniques are investigated in the light of different datasets. Comparison of the proposed SPPF with the original PSO-FCM shows the promise of this spatial-spectral BS solution.

Index Terms—Spatial Spectral Clustering, Band Selection, Hyperspectral, HSI, Superpixel, PSO-FCM

I. INTRODUCTION

Remote sensing using hyperspectral imaging (HSI), captures the reflectance at different wavelengths on the earth's surface, facilitating important applications such as the monitoring of environmental change [1] and the conduction of land cover classification [2]. To address the high dimensionality challenge inherent in HSI, reducing the dimensionality, through the selection of bands that exhibit high separability is necessary [3].

Superpixel-enhanced PSO-FCM (SPPF) proposed herein, builds on PSO-FCM [4] — a hybrid

of Particle Swarm Optimization and Fuzzy C-Means clustering; and proposes the integration of spatial information to achieve a spectral-spatial Band Selection method (BS). In SPPF, superpixels replace the pixels of the PSO-FCM and a superpixel preprocessing step is added. To optimise SPPF for given datasets, various superpixel enhanced distance metrics are proposed and applied as part of the objective function. Various superpixel-adapted cluster selection methods are also investigated in the light of such distance metrics. Experiments are conducted on the Indian Pines (IPDs), Salina (Sds) and Pavia Center (PCds) datasets.

The motivations behind this work together with a brief introduction to the PSO-FCM model is presented in section II. The proposed SPPF model are presented in section III. Experiments and their evaluation may be found in section IV, the conclusion is presented in section V and the source code is available in [5].

II. MOTIVATION

Dimensionality reduction methods can be divided into two categories: those based on feature extraction e.g. principle component analysis; and those based on BS. A challenge with feature extraction is preservation of the information or physical meaning of the original band values [6]. In recent years numerous BS methods have been proposed to address this challenge [2]. Supervised BS approaches

can deliver high classification accuracy but rely on prior ground information - see [7] for an overview. Unsupervised techniques are often preferable for BS due to the scarcity or absence of labelled information [2]. Information Theoretical (IT) approaches as band quality measures are commonly applied and are often used in conjunction with different search algorithms, such as PSO [8] and NSGA-II [9].

There are two main IT approaches: a) entropy of selected bands [4] and b) redundancy and mutual information between bands [10]. For distance measures, standard mutual information approaches are not effective [11]. Solutions include normalised mutual information [10] and the use of disjoint information [11].

Many recent BS approaches incorporate ranking [12] or clustering [11]. Ranking-based techniques score each individual band, displaying a high degree of stability but disregarding correlation information between them. In contrast, clustering-based techniques select bands with lower correlation and less redundant information [8] — k -means clustering algorithm is commonly applied. The cluster selection mechanism is a general challenge for BS. Other approaches include highest average similarity [13]; an iterative similarity approach [11] and ranking combined with clustering [14].

To address the challenge of overlapping band characteristics, soft clustering methods such as Fuzzy C-means clustering [15] are needed. PSO-FCM employs a PSO-based optimisation with FCM as the objective function. However, FCM clustering is sensitive to initial conditions and noisy bands [4]. However, bands known to be uninformative or noisy can be removed *a priori*; to improve performance [16]. However, the performance impact is not commonly addressed.

Supapixel (Sp) segmentation has been utilized effectively in a broad range of applications [17]. Their ability to adapt to the spatial structure of objects is valuable in HSI data analysis [18]. However, most works focus on utilising Sp directly rather than through BS methods. SLIC [14] is a gradient-based method that provides good compaction and regularity [14] but requires adaptation to adjust to HSI. Examples of Sp based unsupervised BS may

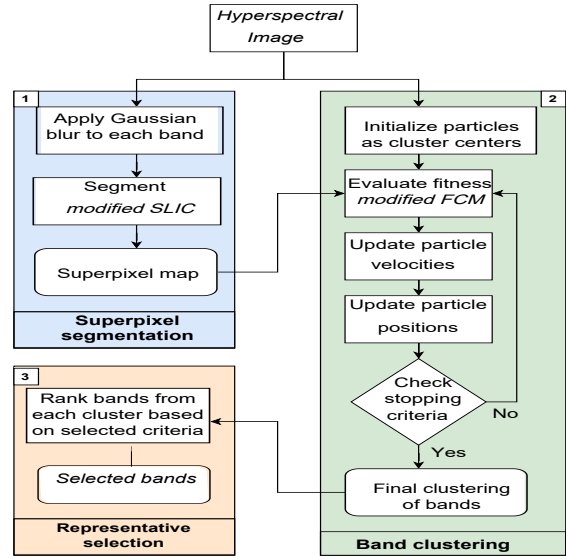


Figure 1: SPPF Architecture

be seen in [14], [19], [20].

Particle Swarm Optimisation (PSO) has gained recognition in BS, due to its ability to navigate large solution spaces effectively [21] and its insensitivity to initialisation. A 2-level PSO approach may be seen in [22] and a GA/PSO hybrid may be seen in [23]. As stated, the work herein extends the PSO approach in [4]. To date, most BS works, including PSO-FCM, consider only the HSI's spectral properties, ignoring spatial properties [8].

III. SUPERPIXEL-ENHANCED PSO-FCM

As shown in Fig. 1, the SPPF model operates in three distinct steps, each contributing to the overall objective of identifying and selecting the most relevant bands for classification. These include: Supapixel Pre-processing (blue) that produces the Superpixel map; a modified PSO-FCM (green) for band clustering and a cluster selection process (orange). Since the work is focused on band selection and not the classifier, the selected bands are fed into a standard Support Vector Machine (SVM) classifier.

1) *Supapixel Pre-processing*: The supapixel pre-processing steps (box 1) involve a) applying Gaussian Blur to every band to reduce high frequency noise which can cause the segmentation

process to produce irregular and fragmented regions and b) superpixel segmentation through the application of a modified version of the Simple Linear Iterative Clustering (SLIC) method [24] - a k-means clustering approach to superpixel creation

The modified SLIC algorithm considers the spectral intensity, rather than the CIELAB colour space, for each pixel in each spectral band. The distance in spectral intensity between pixels i and j is defined as:

$$d_{si}(i, j) = \sum_b (x(i) - x(j))^2 \quad (1)$$

$$d(i, j) = d_{si}(i, j) + w_{sp}d_{sp}(i, j) \quad (2)$$

where b is the total number of bands in the HSI, and $x(i)$ and $x(j)$ are the intensity values of pixel i and j respectively in that particular band, w_{sp} is the spatial weight, and $d_{sp}(i, j)$ is the ED in the spatial dimension.

2) *Band clustering*: The band clustering process (box 2); involves the PSO algorithm that produces n optimal spectral bands as the cluster centroids. The FCM objective function of PSO-FCM is expressed as follows:

$$\min J = \sum_{i=1}^C J_i = \sum_{i=1}^C \sum_{j=1}^N u_{ij}^m d_{ij}^2, \quad (3)$$

where C is the number of cluster centers, N the total number of bands, m the fuzzy exponent, d_{ij} the distance between band i and j . Finally, u_{ij} is the fuzzy membership of band j in cluster i , calculated as:

$$u_{ij} = \frac{\sum_{k=1}^C d_{kj}^{\frac{2}{m-1}}}{d_{ij}^{\frac{2}{m-1}}}. \quad (4)$$

Unlike PSO-FCM, SPPF employs a continuous search space, enabling exploitation of the neighbouring band's spectral similarities. To evaluate a cluster centroid vector CC_i , each dimension of $x-i$ is rounded, rather than the probability distribution of PSO-FCM. Further, SPPF employs a repair mechanism — randomly selected bands, if FCM selects duplicate bands.

Various distance metrics may be applied to exploit the local spatial context provided by the superpixel map when clustering the bands. Herein, the

Superpixel Euclidean Distance (Sp-ED), the Superpixel KL-distance (Sp-KL), the Superpixel Disjoint information (Sp-DI) and the Superpixel-level KL Divergence-L1Norm (Sp-KL-L1) are investigated. More detailed implementation descriptions for these refined metrics are available in [25].

The *Sp-ED* considers the ED between the aggregated values (mean) of the superpixels, rather than individual pixels (as applied in PSO-FCM), aiming to reduce the computational complexity compared, whilst potentially resulting in higher overhead complexity.

Sp-KL quantifies the difference in entropies between spectral bands, promoting the segregation of bands with distinct entropy profiles into separate clusters. To this end normalized grayscale histograms are calculated to obtain the probability distributions of each band, and applied to calculate the Sp-KL between band i and band j

The *Sp-DI* is a modified version of the DI [11]. Sp-DI, similar to Sp-KL is derived from the joint entropy and mutual information of two bands. However, it is less sensitive to differences between small probabilities whilst incurring higher overhead.

The *Sp-KL-L1*, adapted from [14] applies the L1-norm to generate the KL-L1 distance matrix for each Sp. Such distance matrices are stacked together to form a 3D distance cube, $B \times B \times N$, where B is the number of bands and N is the number of Sps. As such, this metric calculates the distance within, and between, each Sp, characterising the Sp structure.

3) *Representative Selection*: For step 3, various selection methods were considered: Highest Entropy (HE); Central Tendency (CT); Hybrid Weighted-sun Criterion (WS) and Hybrid Ranking Critereon (Rank).

HE measures the randomness (information) present in the spectral band, as applied in PSO-FCM. Since similarity is not considered, redundant bands (containing overlapping or highly correlated information) may still be selected. In *CT*, the objective is to represent the majority of the bands within a cluster. However, the band closest to the mean may not necessarily contain as much information as other bands thus only partially capturing the

Table I: Datasets

Dataset	Dimensions	# Bands	Sensor	# Classes
IPds	145x145	220	AVIRIS	16
Sds	512x217	220	AVIRIS	16
PCds	715x1096	102	ROSIS	9

Table II: Hyperparameters for SPPF

Parameter	Description	Value
pop_size	Population size	100
max_iter	Maximum iterations	100
w	Inertia weight	0.7
c_1	Cognitive component	1.0
c_2	Social component	1.0
v_{max}	Max. velocity fraction	0.1
m	Fuzzy exponent	2.0
w_e	Entropy weight	0.35
w_{ct}	Central Tendency weight	1.0

range of information present within the cluster. WS combines HE and CT through a weighted selection and $Rank$ is a summation of a band's entropy and its distance to the mean with the lowest rank selected.

IV. EXPERIMENTS AND RESULTS

The experiments conducted in this work are run on a 6-core 3.2GHz CPU with 32GB of RAM.

1) *Datasets*: The SPPF model's ability to perform well under different spatial, spectral, and geographical conditions were investigated through three well-known hyperspectral datasets that have established benchmarks for classification tasks [26], recorded by two separate sensors in three distinct locations - see Table I. Two artifacts, namely the HSI and the ground truth, are included in each dataset. The HSI is a tensor representing the irradiance values (spectral intensity) detected at the sensor's focal point.

2) *Hyperparameters*: A combination of grid search with trial and error, resulted in the hyperparameters for SPPF — see Table II. Particular emphasis was placed on tuning the weight parameters w_e and w_{ct} , critical to the output from the BS model. To achieve segmentation quality and efficiency, the optimal number of superpixels largely depends on the spatial dimension of the image. Thus preliminary testing provided a recommended average Sp size that was applied to provide the number

Table III: Hyperparameters for Sp segmentation

Dataset	σ	# Superpixels	w_{sp}
IPds	0.5	400	1,000
Sds	0.5	600	1,000
PCds	0.5	900	10,000

Table IV: Hyperparameters for PSO-FCM [4]

Parameter	Description	Value
pop_size	Population size	100
m	Fuzzy exponent	2.0
w	Inertia weight	0.6
c	Cognitive and Social weight	1.75

of Sps needed — see Table III. Further, through a combination of quantitative and qualitative analysis, the spatial weight w_{sp} , and standard deviation for the Gaussian blur σ , were determined. The original PSO-FCM hyperparameters are presented in Table IV.

3) *Classification*: In this study, only the spectral dimension is considered for classification to guarantee a fair comparison to PSO-FCM. The radial basis function is employed as the kernel, and the kernel parameters C and γ are established using grid search and five-fold cross-validation of the training set, as recommended in [27]. This parameter search is repeated for every band subset classified, to ensure a fair comparison. Due to a large discrepancy in available classes for the different samples, each class is split individually before normalising the feature values, guaranteeing the same percentage of training samples for each class - see Table V.

Two accuracy-based metrics are applied for classification performance evaluation : Overall Accuracy (OA) and Average Accuracy (AA). Further, McNemar's test is employed to verify statistical significance. Notably, McNemar's $|Z|$ value of 1.96 or higher denotes a statistically significant difference at a confidence level of 95% or higher ($p \leq 0.05$).

Table V: Training statistics

Dataset	Samples	Labeled samples	Training %
IPds	21,025	10,249	10%
Sds	111,104	54,129	2%
PCds	783,640	148,152	1%

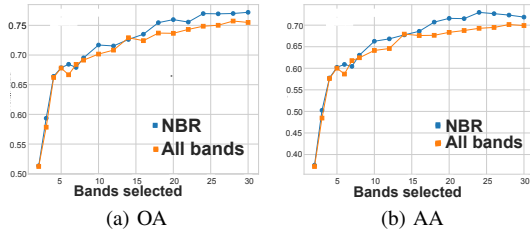


Figure 2: Exp1. NBR performance on IPds

Table VI: Exp1. Specifications

Dataset	IPds, IPds(NBR), Sds, Sds(NBR)
Distance metric	Sp Euclidean distance
Cluster selection	Highest entropy

A. Experiments

For the results presented, each experiment is conducted for 10 runs applying the SPPF model; the stopping criteria is 100 generations or 15 generations without improvement and the number of classification runs are 10, unless otherwise stated.

1) *Exp1: Pre-removal of noisy bands*: Table VI provides the experiment specifications for the pre-removal of noisy bands (NBR) and the results on IPds are shown in Fig. 2. As shown, for both OA and AA, for a lower number of bands selected ($n \leq 15$), the results are inconclusive. However, for Sds, the disadvantage of NBR was clearer — pulling down the performance over all runs. However, there was a positive advantage on IDds and Sds when $n > 15$ — see Table VII. Inclusion or omission of a singular high-entropy band to a cluster has a more significant impact when n is low. NBR is implemented in further experiments.

2) *Exp2. Cluster representative selection*: The optimal cluster representative selection method for a given distance metric was the focus of Exp2 where the setup is given in Table VIII. For the distance

Table VII: Exp1. NBR effect on performance

Dataset	All runs				Runs with $n \geq 15$			
	OA	OA SD	AA	AA SD	OA	OA SD	AA	AA SD
IPds	0.698	0.013	0.630	0.020	0.744	0.015	0.689	0.021
IPds(NBR)	0.708	0.014	0.646	0.021	0.761	0.014	0.715	0.019
Sds	0.879	0.006	0.905	0.009	0.907	0.003	0.943	0.003
Sds(NBR)	0.877	0.005	0.899	0.007	0.908	0.002	0.944	0.002

Table VIII: Exp2. Specifications

Dataset	IPds, Sds, PCds
Distance metric	Sp-ED, Sp-KL, Sp-DI, Sp-KL-L1
Cluster selection	HE, CT, WS, Rank

Table IX: Exp2a. Sp Euclidean Distance

	IPds		Sds		PCds	
	OA	AA	OA	AA	OA	AA
HE	0.758	0.706	0.903	0.939	0.976	0.923
CT	0.779	0.736	0.909	0.944	0.976	0.921
WS	0.778	0.734	0.908	0.944	0.976	0.922
Rank	0.771	0.720	0.907	0.943	0.975	0.921

metric Sp-ED — see results in Table IX; there is a slight advantage overall with CT over WS and thus CT is chosen for this distance metric. Interestingly all selection methods performed similar on PCds — even HE despite it being poorer on the other datasets.

When Sp-KL is applied, CT struggles compared to WS on the IPds whilst WS performs on par or better than other metrics across the 3 datasets — see Table X. Interestingly, the difference between the best and second-best criterion for the IPds was larger than any other distance metric, highlighting the unique advantage of combining the Sp-KL and WS.

Again in the Sp-DI results shown in Table XI, a close competition between CT and WS may be observed. HE shows poorer performance, as seen in Sp-KL (Table X). Despite Rank's poorer performance on IPds and Sds, it again performs on par with the other selection methods on the PCds, suggesting that Rank may be a suitable for certain datasets and thus should not be discounted. However, CT was chosen for Sp-DI.

Table XII provides performance results for apply-

Table X: Exp2b. Sp KL-Divergence

	Indian Pines		Salinas		Pavia Center	
	OA	AA	OA	AA	OA	AA
HE	0.745	0.689	0.899	0.936	0.974	0.918
CT	0.758	0.710	0.902	0.939	0.975	0.920
WS	0.765	0.716	0.902	0.939	0.975	0.920
Rank	0.754	0.697	0.897	0.934	0.975	0.921

Table XI: Exp2c. Sp Disjoint Information

	IPds		Sds		PCds	
	OA	AA	OA	AA	OA	AA
HE	0.734	0.678	0.896	0.930	0.970	0.905
CT	0.742	0.679	0.899	0.936	0.975	0.919
WS	0.740	0.679	0.900	0.937	0.975	0.918
Rank	0.733	0.665	0.893	0.929	0.975	0.919

Table XII: Exp2d. Sp KL Divergence-L1Norm

	IPds		Sds		PCds	
	OA	AA	OA	AA	OA	AA
HE	0.767	0.717	0.901	0.936	0.975	0.921
CT	0.782	0.741	0.907	0.942	0.976	0.923
WS	0.779	0.735	0.905	0.941	0.976	0.924
Rank	0.780	0.737	0.903	0.938	0.976	0.923

ing Sp-KL-L1. CT shows strong performance across all 3 datasets although there is a slight advantage for WS for PCds.

It is clear that CT and WS exhibit good performance across all distance metrics, suggesting the importance of a similarity within the metric.

3) *Exp3. Distance Measures*: The goal of these experiments is to study how the Sp enhanced distance metrics affect the classification performance and efficiency.

Table XIII: Exp3. Specifications

Dataset	IPds, Sds, PCds
Distance metric	Sp-ED, Sp-KL, Sp-DI, Sp-KL-L1

When IPds is considered, Fig. 3 shows that the information theoretic approaches struggle when $n = 5$ but with a rapid improvement from $5 > n < 15$. High variability in the few bands selected effects the entropy of such measures. Further, outlier bands with high entropy can have an adverse affect on the representative band selection when applying CT or WS. Fig. 4 highlights a 10 band and 15 band case for Sp-kl-L1 with the bands selected (green) and their respective entropy values (blue). It may be noted that in the case of $n = 15$, the bands selected are more evenly distributed, reducing the effect of outlier bands.

As shown in Fig. 3, Sp-KL-L1 shows superior and more stable performance compared to Sp-KL,

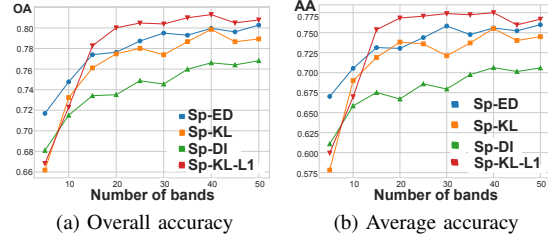


Figure 3: Exp3a. Accuracy on IPds

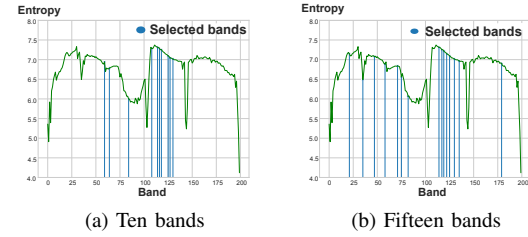


Figure 4: Exp3a. BS applying SP-KL-L1 to IPds

highlighting the advantage of considering the relative entropy of the SPs between bands rather than the aggregated mean values. Further, it is noted that Sp-DI is outperformed by all other distance metrics when the superpixel's mean values are too divergent.

The performance results of SPPF on Sds — see Fig. 5; highlight similar effects for low n . However, at higher n , an improved classification accuracy and less performance difference may be seen across the distance metrics, compared to IPds. However, given the spatial resolution of the Sds compared to the IPds this is to be expected. It should be noted that Sp-ED performs on par with Sp-KL-L1, superior to Sp-DI and Sp-KL. Sp mean values would seem to be representative of the SP pixels, making divergence measures less effective.

The findings for PCds, presented in Fig. 6, reveal that all distance metrics employing mean values (Sp-ED, Sp-KL, Sp-DI) show high and similar accuracy for $n \geq 20$, as expected for PCds — with its smaller number of classes combined with high spatial resolution. Further, Sp-KL-L1 shows a significant peak at $n = 30$. Fig. 7 (a) shows that for Sp-KL-L1, the average entropy of bands selected is high compared to Sp-ED with the effect reducing as

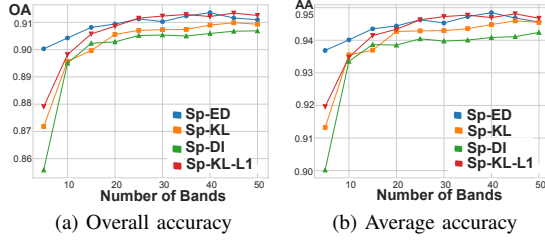


Figure 5: Exp3b. Accuracy on Sds

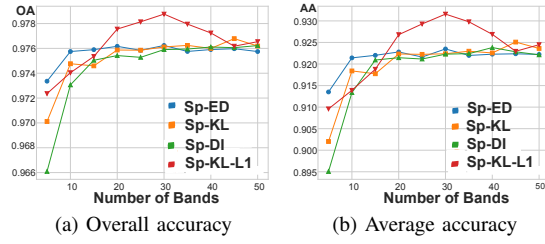


Figure 6: Exp3c. Accuracy on PCds

n increases. In (b), Sp-ED (blue), achieves a more even dispersion of bands, including low entropy bands. However, for $n > 30$, the fall in Sp-KL-L1 performance may be attributed to the high number of bands selected relative to the available bands - roughly half of the available bands.

To consider efficiency, Fig. 8 (a) shows the mean runtime for a single iteration when selecting 20 bands. Interestingly, the runtime for the PCds (in green) is substantially lower than the other datasets, even though it has a much higher spatial resolution and a higher number of Sps. This is again likely due to the low number of available bands. However, from Fig. 8 (b), the overhead calculations of the PCds are higher for all distance metrics and substantial when Sp-KL-L1 metric is applied. On the

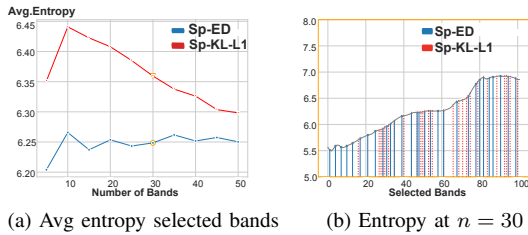


Figure 7: Exp3d. Comparison of Entropy on PCds

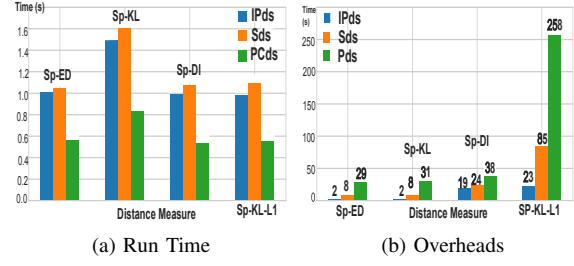


Figure 8: Exp3e. SPPF Efficiency

Table XIV: Exp4. Specifications

Model	SPPF, PSO-FCM
Dataset	IPds, Sds, PCds

other hand, the overhead of the Sp-ED is relatively low and the mean runtime competes with the Sp-DI. As such, the Sp-ED seems more efficient whilst the SP-KL-L1 shows best performance and given the importance of efficiency for Band selection, Sp-ED is chosen.

4) *Exp 4: SPPF Comparison with PSO-FCM:* SPPF is a Sp extension of PSO-FCM and the effect of this extension on performance and efficiency is investigated with the specifications given in Table XIV. As shown in Fig. 9, the SPPF model outperforms PSO-FCM on IPds with ($n \geq 15$); both in terms of higher accuracy and stability — standard deviation highlighted in the shaded regions. However, when $n < 15$ the PSO-FCM outperforms SPPF and SPPF exhibits low stability, as expected from Exp3. However, SPPF outperforms PSO-FCM, even with lower n and with little standard deviation on Sds - see Fig. 10. On PCds, despite strong classification results, each methods excels at different levels of n but with substantial standard deviation. Results confirm that both SPPF and PSO-FCM struggle to distinguish objects when parts of that area of the image are missing, as in PCds.

Table XV provides the McNemar's test results on the performance of SPPF versus PSO-FCM, based on 10 trials for each combination of dataset and bands selected. It should be noted that values over 1.96 imply that SPPF outperforms PSO-FCM with

Table XV: Exp4d. McNemar's test

ds	n	Z ₀	Z ₁	Z ₂	Z ₃	Z ₄	Z ₅	Z ₆	Z ₇	Z ₈	Z ₉
IPds	20	5.08	13.99	6.93	10.75	3.26	10.07	14.96	14.0	6.79	8.59
	30	9.01	10.67	7.62	5.27	6.56	6.62	8.73	7.86	10.67	4.94
	40	10.12	-0.80	0.98	12.76	3.17	6.23	10.47	7.03	6.38	9.75
Sds	20	4.68	2.51	4.47	1.04	0.81	1.52	2.18	6.49	0.82	3.91
	30	5.67	-1.18	-1.27	2.56	-1.41	7.89	2.01	3.67	3.69	1.85
	40	-2.57	0.41	3.74	-5.36	-0.37	2.70	-0.92	4.05	-0.66	2.96
PCds	20	1.89	0.40	4.59	3.55	0.42	3.47	-0.93	-3.26	-3.09	4.58
	30	-0.27	5.59	0.25	0.01	-1.35	3.53	0.67	3.37	12.87	0.72
	40	7.22	-2.24	-3.62	8.17	-6.33	-1.58	0.71	3.88	-0.62	2.28

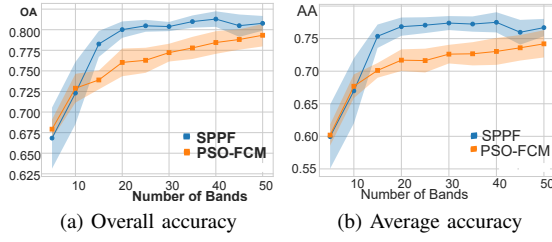


Figure 9: Exp4a. Accuracy on IPds

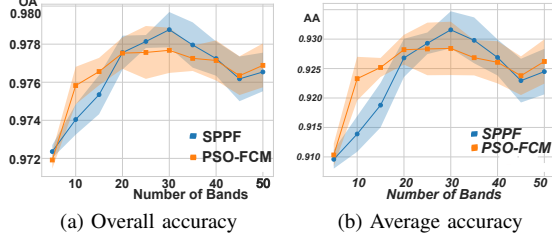


Figure 11: Exp4c. Accuracy on PCds

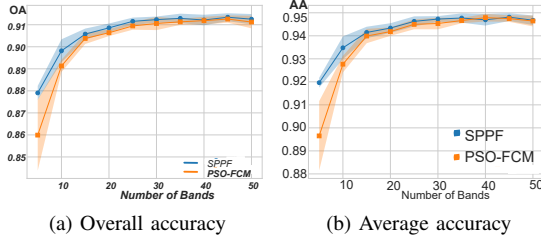


Figure 10: Exp4b. Accuracy on Sds

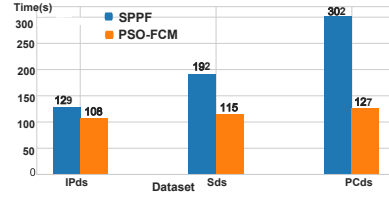


Figure 12: Exp4d: Efficiency

95% confidence and values between 1.96 and -1.96 are inconclusive.

SPPF significantly outperforms PSO-FCM for IPds, particularly for $n = 20, 30$. For Sds, the results are more varied and inconclusive. For PCds the results show a high variability, alternating in their superior performance across different runs. However, the overall trend (90 trials) is that SPPF significantly outperforms PSO-FCM on 56 trials; is significantly outperformed on 7 and for 30 trials there is no significant difference.

From Fig. 12 with $n = 20$ bands, and 100 runs, where IPds then Sds and then PCds are presented, it is clear that PSO-FCM is on average more efficient across all 3 datasets with the difference more significant for datasets with a higher number of pixels,

despite the introduction of Sps. However, this was not unexpected, given the priority of classification accuracy over efficiency from the results in Exp3.

V. CONCLUSION

The work herein considered different ways to incorporate spectral-spatial information into the PSO-FCM algorithm through the introduction of Sps. The results highlighted the benefits of removing noisy bands, different distance measures and different cluster-representative selection techniques. Further, SPPF enhanced performance over PSO-FCM but was less efficient. Future work will focus on improving efficiency, testing on further datasets and comparison with other State-of-the-Art BS methods, particularly with respect to scalability.

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